

THE ISOMERISM MAP

ISOMERS

same molecular formula · different structure

STRUCTURAL

different atom CONNECTIVITY

Chain

Position

Functional

Metamerism

Tautomerism

Ring-chain

STEREO

same connectivity · different 3-D shape

Geometrical

cis-trans / E-Z

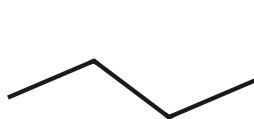
Optical

chiral · R/S

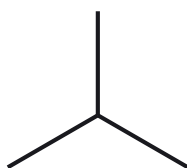
Conformational

free rotation

① CHAIN (skeletal)



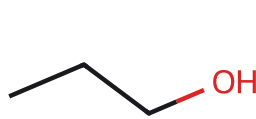
n-butane



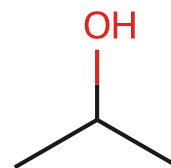
isobutane

different carbon skeleton C_4H_{10}

② POSITION



propan-1-ol



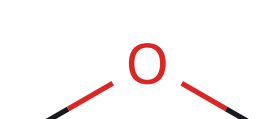
propan-2-ol

same group, new position C_3H_8O

③ FUNCTIONAL



ethanol · alcohol



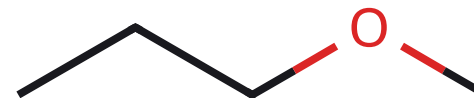
dimethyl ether

different functional group C_2H_6O

⑥ METAMERISM



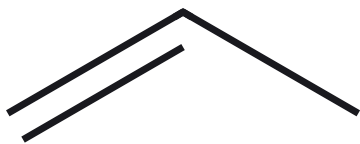
diethyl ether



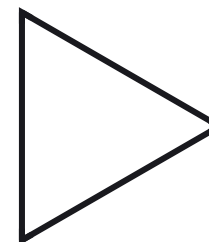
methyl propyl ether

unequal alkyls around O/N/S $C_4H_{10}O$

⑦ RING-CHAIN



propene



cyclopropane

open chain $\rightleftharpoons$ ring C_3H_6

⑤ TAUTOMERISM (keto $\rightleftharpoons$ enol)



OTHER FUNCTIONAL-GROUP ISOMER PAIRS

aldehyde $\rightleftharpoons$ ketone

acid $\rightleftharpoons$ ester

alcohol $\rightleftharpoons$ ether

$-CN \rightleftharpoons -NC$

$-NO_2 \rightleftharpoons -ONO$

$1^\circ \cdot 2^\circ \cdot 3^\circ$ amines

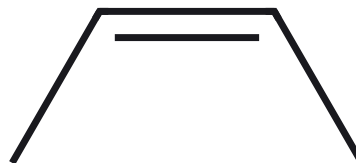
WHEN DOES IT OCCUR?

✓ restricted rotation (C=C or ring)

✓ each sp^2 C has 2 different groups

✗ any C with 2 same groups → none

cis vs trans



cis · (Z)-but-2-ene

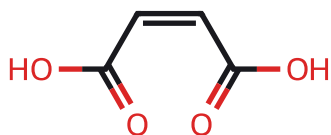


trans · (E)-but-2-ene

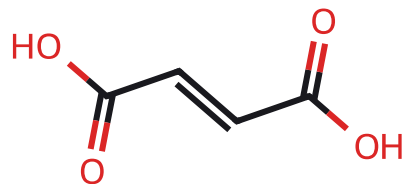
trans: ↑ m.p., stabler

cis: ↑ dipole

MALEIC vs FUMARIC



maleic acid



fumaric acid

	MALEIC (cis)	FUMARIC (trans)
m.p.	130 °C (low)	287 °C (high)
dipole	polar	non-polar
anhydride	easy on heating	hard
K_{a1}	higher	lower
K_{a2}	lower	higher
stability	less	more

ALSO SHOWS G.I.



syn (Z)-oxime



anti (E)-oxime

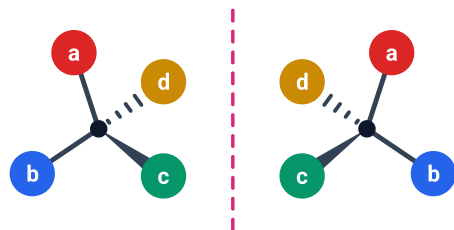
C=N oximes syn/anti

N=N azo

rings

Ring C=C: C_3 – C_7 exist as **cis only** (ring too small for trans); **trans** from C_8 (trans-cyclooctene) onward

CHIRALITY



▲ bold wedge = toward you | hashed = behind

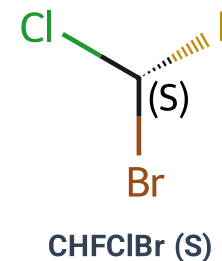
non-superimposable mirror images = **ENANTIOMERS**

chiral C = **4 different** groups (asymmetric C)

ASSIGN R / S CONFIGURATION



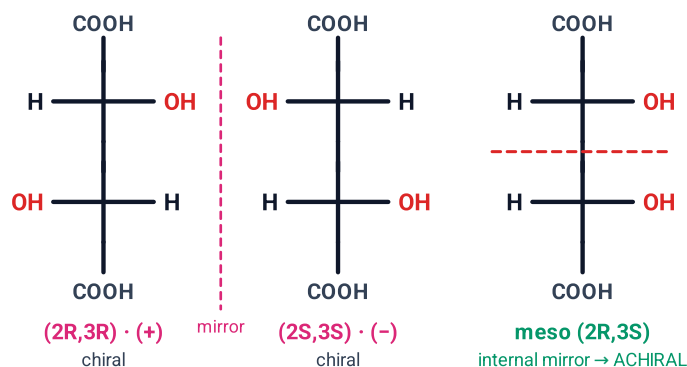
lowest (4) points back
1→2→3 clockwise = R
anticlockwise = S
(4 in front → reverse)



CIP priority

- 1 · higher atomic number
- 2 · tie → next atom, 1st difference
- 3 · isotope: higher mass
- 4 · C=O counts as C bonded to (O,O)

ENANTIOMER · DIASTEREOMER · MESO



2 enantiomers (optically active) + 1 meso (inactive) = 3 stereoisomers

enantiomers: mirror; same props except ± rotation

diastereomers: non-mirror; different props

meso: internal mirror → achiral

Fischer moves: 1 interchange = R↔S (inversion) · 2 interchanges = retention · rotate 90° → inverts · 180° → retains

Anti addition (Br₂): cis-alkene → (±) racemic · trans-alkene → meso (syn addition reverses)

COUNTING STEREOISOMERS

molecule (n = chiral C)	total forms	optically active	meso
unsymmetrical	2^n	2^n	0
symmetrical, n even	$2^{n-1} + 2^{(n/2)-1}$	2^{n-1}	$2^{(n/2)-1}$
symmetrical, n odd	2^{n-1}	$2^{n-1} - 2^{(n-1)/2}$	$2^{(n-1)/2}$

e.g. tartaric acid (n = 2, symmetric) → **3** forms = 2 active (d, l) + 1 meso

SPECIFIC ROTATION & ee

$$[\alpha] = \alpha / (l \cdot c)$$

α = observed rotation (°) · l = path length (dm) · c = concentration (g/mL)

(+) / d = dextro (clockwise)

(-) / l = laevo (anticlockwise)

$$\text{optical purity} = \text{ee \%} = (\text{observed } [\alpha] \div \text{pure } [\alpha]) \times 100 = |\%R - \%S|$$

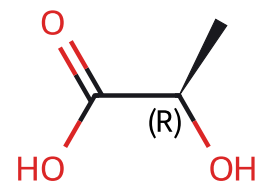
racemate ($\pm$) 50:50 → inactive · resolve via diastereomeric salts / enzymes · %R = 50 + ee/2

D/L vs d/l vs R/S

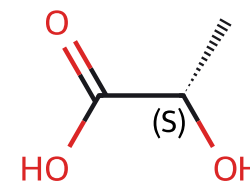
label	meaning
R / S	absolute configuration (CIP)
D / L	relative config vs glyceraldehyde
+ / - (d/l)	measured direction of rotation

No D/L ↔ +/- correlation · Fischer: 90° inverts, 180° retains

ENANTIOMER PAIR



(R)-lactic acid



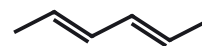
(S)-lactic acid

equal & opposite $[\alpha]$; identical m.p. / b.p.

COUNTING GEOMETRICAL ISOMERS (POLYENES)

polyene (N = C=C w/ cis-trans)	geometrical isomers
unsymmetrical	2^N
symmetrical, N even	$2^{N-1} + 2^{(N/2)-1}$
symmetrical, N odd	$2^{N-1} + 2^{(N-1)/2}$

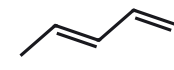
2,4-hexadiene · symmetric · N = 2 → 3 isomers



trans, trans



cis, cis



trans, cis

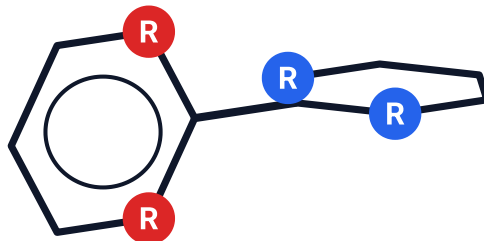
CHIRALITY WITHOUT A CHIRAL CARBON (axial)



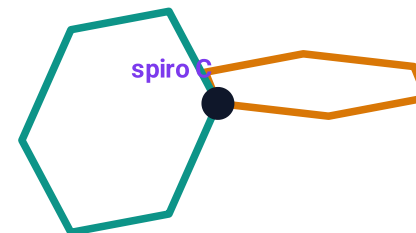
allene: two end planes are $\perp$

CUMULENES ($C=C=\dots=C$)

- EVEN number of $C=C$ (2,4,...) $\rightarrow \perp$ ends $\rightarrow$ OPTICAL
- ODD number of $C=C$ (1,3,5,...) $\rightarrow$ coplanar $\rightarrow$ GEOMETRICAL

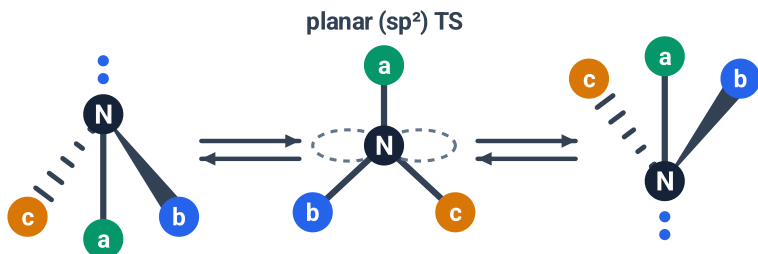


2,2',6,6'-tetra-substituted biphenyl
bulky ortho R $\rightarrow$ restricted rotation
ATROPISOMERS (axial chirality)



SPIRO – two rings share ONE sp^3 C
the two rings lie in $\perp$ planes
chiral (axial) when the ends differ

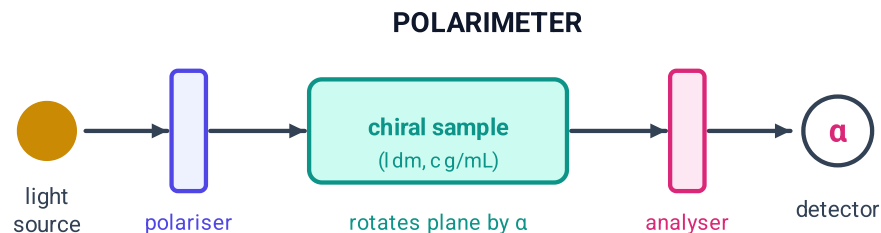
AMINE INVERSION – a 'stereocentre' that can't be resolved



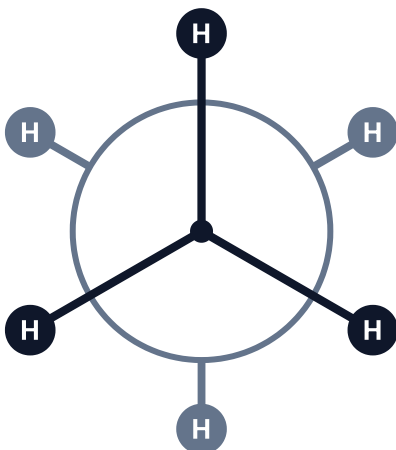
nitrogen umbrella inversion – fast at room temp
the two mirror forms interconvert $\rightarrow$ OPTICALLY INACTIVE

N (3 different groups + lone pair) looks chiral, but inversion is too fast at RT $\rightarrow$ optically inactive · quaternary N^+ (4 diff.) is chiral

POLARIMETER – how $[\alpha]$ is measured

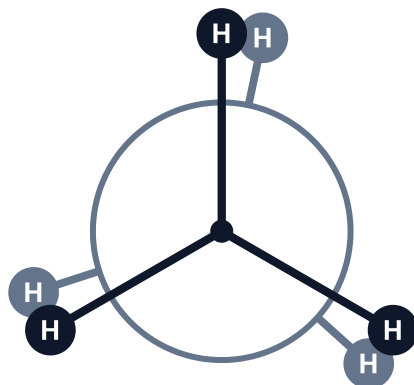


ETHANE



STAGGERED

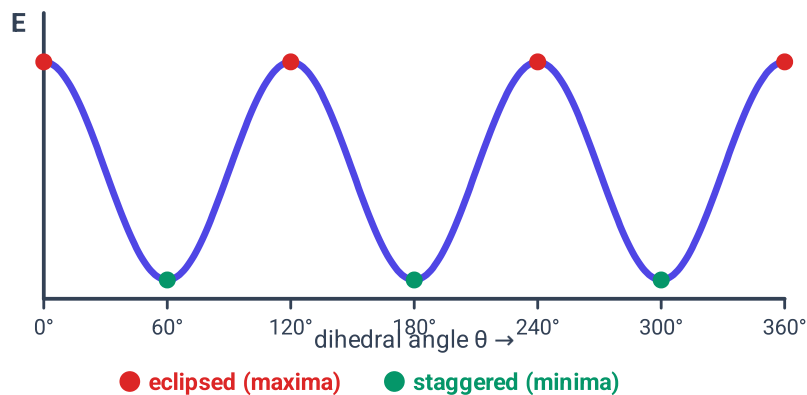
$\theta = 60^\circ$ · lowest energy



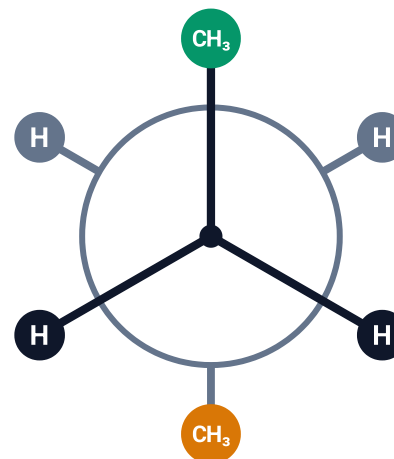
ECLIPSED

$\theta = 0^\circ$ · highest energy

ETHANE — barrier ≈ 12.5 kJ/mol

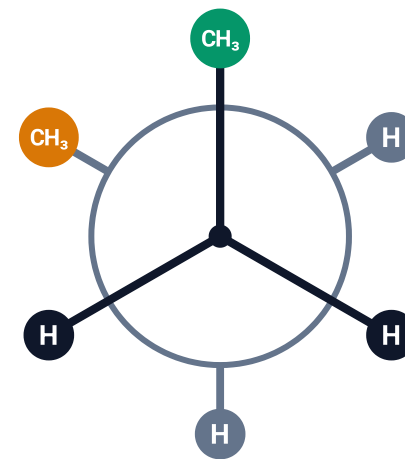


n-BUTANE (rotate C2–C3)



ANTI (180°)

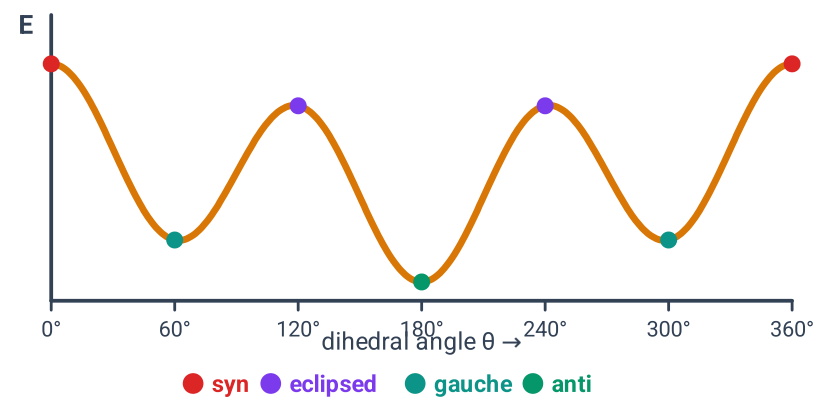
most stable



GAUCHE (60°)

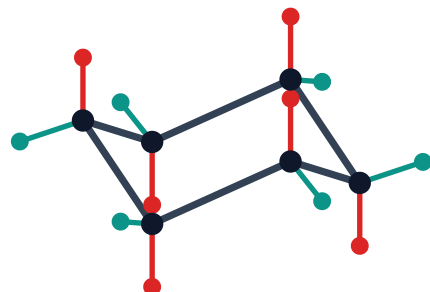
+3.8 kJ/mol

n-BUTANE (rotate C2–C3)

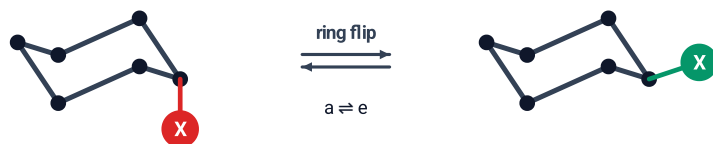
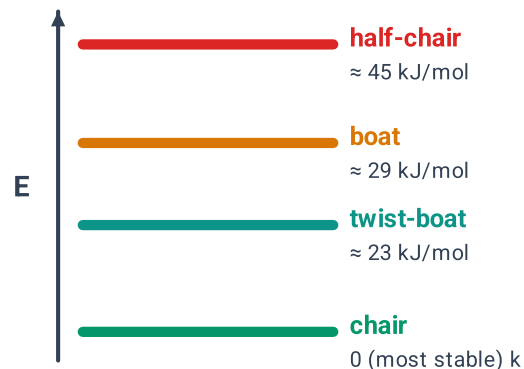


anti > gauche > eclipsed > fully-eclipsed (syn)

CYCLOHEXANE (chair is king)

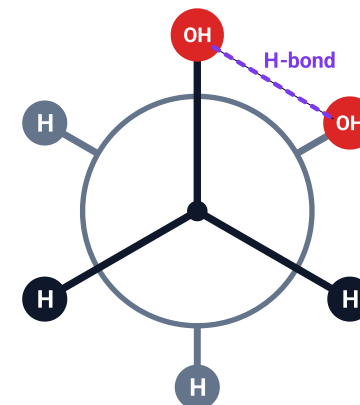


● axial (a) – vertical
● equatorial (e) – outward



bulky group prefers EQUATORIAL (avoids 1,3-diaxial strain)

GAUCHE EXCEPTION – INTRAMOLECULAR H-BOND



ethylene glycol · GAUCHE

intramolecular $O-H \cdots O \rightarrow$ gauche > anti

also $FCH_2CH_2OH \cdot H_2NCH_2CH_2OH$: H-bond $\rightarrow$ gauche > anti